

The Hidden Cost

Ask AI something simple and the answer shows up in a second, like it cost nothing. Behind that second is one of the most demanding things a computer can do.

What one answer costs

● YOU

What's a good name for a golden retriever puppy?

● AI

A few favorites: Sunny, Biscuit, or Cooper. Want more, or names with a theme?

~40 trillion

CALCULATIONS, FOR THAT ONE REPLY

~50 tokens

×

~400B parameters

×

2 ops each

Every token re-runs the whole network: all of the model's parameters, each doing a couple of operations. That's the loop you watched in Inference, and it is why a one-line answer still adds up to trillions.

HARD TO PICTURE? STACK IT UP

40 trillion dollar bills, end to end, reach past Pluto.

A dollar bill is about six inches long. Lay down one for every calculation in that single answer and the line runs about 6 billion kilometers.

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Earth to Pluto · ~6 billion km of dollar bills

Rough numbers: the exact count depends on the model's size and how long the answer is. The scale is the part that's real.



WHY DATA CENTERS EXIST

No single computer can do that fast, for millions of people at the same time. So AI companies build **data centers**: warehouses packed with thousands of specialized chips, called GPUs, running around the clock.

THE FOOTPRINT

All that hardware draws real electricity, and cooling it takes real water. As more people use AI, that footprint grows. It isn't unique to AI; every big technology has one, from streaming video to the cars outside, and engineers are racing to make the chips far more efficient. But it is real, and most people never picture it.

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Now you know what's behind the magic. An instant answer feels free, but real chips, power, and water are working behind every one. None of this is a reason to feel guilty hitting send. It's a reason to be one of the few people who actually understands the machine you're using.

TRY IT

AI's energy and water footprint gets talked about in extremes. True or false? Decide each, then check your reasoning.